

LLM-Based Structured Extraction from Clinical Dialogue: An Evidence-Grounded Supporting Report

Supporting Report

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Abstract

This supporting report documents, in full technical depth, the design, implementation, verification, and critical appraisal of a study investigating how an open-source large language model can be adapted for structured information extraction from clinical doctor-patient dialogue. Conversational text is informal, multi-turn, and context-dependent, properties that extraction systems built for static documents do not handle well. A custom, human-reviewed, 798-example benchmark was constructed from the MTS-Dialog corpus using Qwen3-8B as an annotator, with two competing automated quality-filtering strategies trialled before a full human review pass. Four research questions were then tested against this benchmark: whether LoRA fine-tuning outperforms zero- and few-shot prompting; whether explicit annotation guidelines embedded in the fine-tuning prompt improve generalisation to clinical specialties unseen during training; which of three structured output representations — JSON function-calling, natural-language labels, or Python dataclass instances — minimises hallucination; and how extraction performance degrades with conversational complexity and specialty unfamiliarity. Naive fine-tuning produced a null result relative to prompting; guideline-conditioned fine-tuning produced a modest but real improvement with reduced hallucination; JSON output decisively minimised hallucination against the alternatives; and a measurable but partly artefactual generalisation gap was observed between seen and unseen specialties. Beyond restating these results, this report additionally provides a mathematical formulation of the extraction and scoring task, a full account of the project's life cycle and verification methods, a justified inventory of every tool used, and an honest critical appraisal — including two worked case studies, a disclosed AI-use statement, and an evidence index linking every reported claim to its source artefact.

Keywords

Large Language Models; Information Extraction; LoRA Fine-Tuning; Conversational AI; Clinical Natural Language Processing; Hallucination; Reproducibility.

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Chapter 1 — Introduction

1.1 Context and the Extraction Problem

Information extraction (IE) — the transformation of unstructured text into structured, machine-actionable records — has traditionally assumed static, well-formed input: news articles, scientific abstracts, legal filings. Conversational data violates every one of these assumptions. Utterances are informal, incomplete, spread across multiple turns between speakers, and frequently rely on implicit context — a symptom mentioned three turns earlier, referenced only as “it”; a question asked and then answered with a bare “no”. This gap matters increasingly as organisations accumulate large volumes of conversational data — clinical encounters, customer-service transcripts, voice-assistant logs — that must be mined for structured, actionable content. A hospital needs a structured clinical summary from a doctor-patient dialogue; the same underlying extraction problem recurs, in different clothing, across every domain where humans talk rather than write.

Large language models have demonstrated substantial zero-shot IE capability, but this capability is neither free nor uniform. GPT-3.5 achieves only 18.22 F1 on standard named-entity recognition without task-specific adaptation [3], and zero-shot dialogue-state tracking trails fine-tuned models by 10–20 points on MultiWOZ [6], a benchmark built on the neural dialogue-systems paradigm established by earlier pretrained-language-model approaches [5]. Conversational IE further compounds these weaknesses: existing benchmarks evaluate entity, relation, intent, and event extraction largely in isolation, and none jointly assesses all of these within the same multi-turn conversational context. This project addresses that gap directly by evaluating LLM-based extraction on the highest-stakes conversational setting available within its scope — clinical dialogue — where extraction errors carry real consequences and where the tension between fine-tuning effort, prompt engineering, and hallucination control is most acute.

1.2 Engineering Problem versus Scientific Problem

It is useful to separate what this project had to build from what it set out to discover. The engineering problem is concrete: read a doctor-patient conversation, produce a structured extraction of its clinically relevant content, and score that extraction against a human-verified gold standard. This project implements that loop directly — an annotation pipeline, two competing quality-filtering strategies, a LoRA fine-tuning pipeline, and a three-format evaluation harness (Chapter 4) — and every stage produces an inspectable artefact (a JSON file, a CSV of scores, a saved adapter).

The scientific problem is different, and is the one the four research questions actually target: whether small-scale parameter-efficient fine-tuning is worth its cost relative to prompting alone; whether a specific, low-cost intervention (embedding annotation guidelines in the fine-tuning prompt) measurably improves generalisation to specialties the model has never been fine-tuned on; which structured output representation should be the default choice under a hallucination-sensitive evaluation standard; and how much of an observed generalisation gap is a genuine model limitation versus an artefact of how the benchmark itself is scored. Building a working extraction pipeline was necessary to answer these questions, but it is not, on its own, the contribution — a pipeline that extracts structured facts without establishing whether its adaptation strategy generalises, or whether its apparent gains survive a fair comparison against a strong baseline, would not answer any of them.

1.3 Research Question, Aim and Objectives

The aim of this project is to determine how an open-source LLM can be reliably adapted — through fine-tuning, prompt design, and output-format selection — to extract structured clinical facts from

doctor-patient conversation, and to characterise the conditions under which each adaptation strategy succeeds or fails. Four objectives follow, each mapped directly to one research question addressed in Chapter 6:

Establish whether parameter-efficient LoRA fine-tuning improves on zero-/few-shot prompting alone, on a held-out specialty split (RQ1).

Establish whether explicit annotation guidelines, embedded in the fine-tuning prompt, improve generalisation to specialties unseen during training (RQ2).

Establish which of three structured-output representations — JSON function-calling, natural-language labels, or Python dataclass instances — best balances accuracy against hallucination (RQ3).

Characterise how extraction quality degrades with conversational complexity and specialty familiarity, and how much of any observed gap is genuine versus a scoring artefact (RQ4).

The expected contribution is threefold: a human-verified, difficulty- and specialty-stratified conversational IE benchmark for the clinical domain; a reusable evaluation harness scoring three distinct structured-output formats against a single set of gold labels; and an honest empirical account of where a small-scale, consumer-hardware LoRA fine-tuning pipeline does and does not outperform prompting — including a genuine null result, reported rather than concealed.

1.4 Contributions and Claim Boundaries

Several boundaries are stated here explicitly, in the interest of the same methodological honesty this report returns to throughout. This project makes no claim that its findings generalise beyond a single clinical domain; no claim of superiority over a proprietary model, since no GPT-3.5/4 comparison was run; no claim that its LoRA hyperparameters represent an optimum, since they were adopted from literature defaults rather than searched; and no claim that the resulting system is suitable for autonomous clinical documentation or decision-making — it is positioned strictly as a clinician-facing drafting aid requiring mandatory human review (Chapter 8). Following a supervisor-approved scope review, the project's original plan — spanning at least three domains, proprietary and literature-baseline comparisons, and five research questions — was formally narrowed to a single domain and four research questions (Chapter 7). These boundaries are the single largest limitation of the present work and are revisited critically in Chapter 8, stated here as a scope decision rather than left implicit.

1.5 Report Map

Chapter 2 synthesises the project's literature basis into a critical, per-paper review and positions the present work relative to the gap it leaves open. Chapter 3 defines the extraction task, the three output representations, and every scoring metric mathematically, independent of any particular model or run. Chapter 4 explains the proposed methodology and system design: the end-to-end pipeline architecture, the annotation and guideline prompts actually used, and the model/training configuration. Chapter 5 records the accepted dataset, implementation, and experimental protocol in full detail, including threats to validity stated before any result is presented. Chapter 6 presents results in the order the research questions were posed, interprets them mechanistically, and includes two worked case studies drawn directly from the benchmark. Chapter 7 reconstructs the project as a sequence of decisions, tool choices, and verification steps. Chapter 8 appraises what worked, what did not, ethics, and discloses the AI assistance used in preparing this report. Chapter 9 gives the conclusion and future work. References and two appendices — an evidence index and selected reproducibility notes — close the report.

Chapter 2 — Extended Literature Review

The Research Paper submitted alongside this report compresses its own literature review to approximately one page, as required by the module's writing-style rules for that deliverable. This chapter provides the fuller investigation behind that compressed review: a complete critical treatment of the four core papers underpinning this project's design, the supporting technical and corpus literature it draws on, and a taxonomy that positions all four core papers — and this project — along two axes that matter for conversational clinical extraction: whether the source text is static or conversational, and whether the target schema is closed or open-ended.

2.1 Scope and Taxonomy

Ten sources are referenced throughout this project (this report's References list). Four form the core methodological basis and are reviewed individually below; the remaining six supply the technical foundations (LoRA, QLoRA, the Qwen3 model family) and corpus/context background (MTS-Dialog, and the task-oriented dialogue literature the project's Introduction situates its motivating statistic against). Figure 1 summarises where each core paper sits relative to this project along the two axes just described.

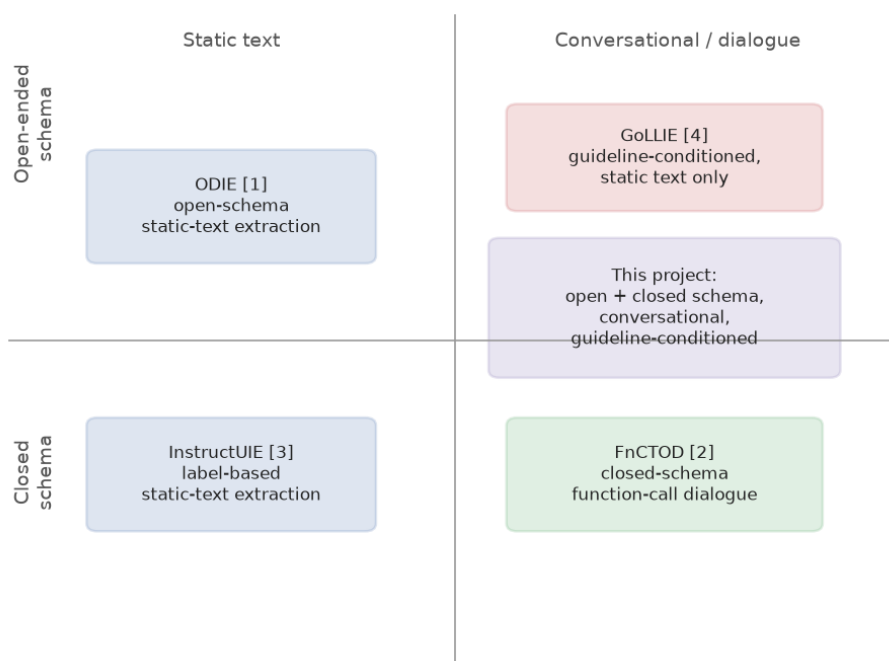


Figure 1. Taxonomy of the four core reviewed papers by source-text type (static vs. conversational) and schema openness (closed vs. open-ended), with this project's position shown for comparison.

Source: Report-only diagram, positions derived from each paper's stated task formulation.

2.2 On-Demand Information Extraction (ODIE) — Jiao et al., EMNLP 2023 [1]

Approach. Jiao et al. introduce On-Demand Information Extraction: a user supplies a natural-language instruction describing what should be extracted, and the model returns the requested content as a structured table, rather than being locked to one fixed extraction schema. The authors construct the InstructIE dataset — 14,579 ChatGPT-generated training pairs spanning 61 domains, plus 150 human-annotated test examples — and fine-tune LLaMA-7B with LoRA in two variants: ODIE-Direct, which

emits the table immediately, and a chain-of-thought variant that reasons before emitting. Evaluation combines Soft-Match F1 (via SentenceBERT similarity, so paraphrases are not penalised as harshly as under exact-match F1), ROUGE-L, and human ratings.

Merit. ODIE-Direct outperforms comparable open-source baselines by roughly 5% and nearly matches GPT-4 despite being two orders of magnitude smaller — a concrete existence proof that a small, LoRA-adapted open model can approach proprietary-model performance on structured extraction from free text, which directly motivated this project's choice to attempt the same with Qwen3-8B rather than assuming a large proprietary model would be required.

Weakness and gap. ODIE operates exclusively on static, single-author documents. Its six-step ChatGPT-based data-generation pipeline has no mechanism for multi-turn context: it does not resolve pronoun or ellipsis reference across turns, does not distinguish speakers, and has no notion of a fact being confirmed versus merely raised and then denied — exactly the properties that make doctor-patient dialogue harder than the news/Wikipedia-style text ODIE was evaluated on.

How this project differs. ODIE's six-step data-generation pipeline is the direct template this project adapted for benchmark construction (Chapter 4), substituting Qwen3-8B for ChatGPT as the annotator and multi-turn doctor-patient transcripts for ODIE's static source documents. Where ODIE's automated filtering step is a single LLM pass, this project trialled two distinct filtering strategies before adding a full human review pass — a more rigorous quality-control chain, made necessary precisely because dialogue introduces the negation and misattribution failure modes that static-text extraction does not.

2.3 Zero-Shot Dialogue State Tracking through Function Calling (FnCTOD) — Li et al., ACL 2024 [2]

Approach. Li et al. reframe dialogue-state tracking as a function-calling problem: each dialogue domain is modelled as a distinct callable function, and the slot values the user has specified become that function's arguments, with the model emitting a JSON function call. A two-step decomposition — select the domain, then generate its arguments — shortens the prompt and improves accuracy. A version fine-tuned on 7,200 dialogues across 36 domains (FnCTOD-LLaMA2-13B) matches ChatGPT's zero-shot DST performance and improves on GPT-4's average Joint Goal Accuracy by 14%.

Merit. FnCTOD is the paper in this review set most directly situated in the conversational domain, and its function-calling framing gives a genuinely dialogue-native precedent for representing extracted state as a structured, machine-parseable call rather than free text — exactly the property this project's RQ3 investigates for a different task.

Weakness and gap. FnCTOD's schema is closed and fixed: each function's argument set is a small, enumerable slot vocabulary (cuisine type, price range, area). This suits task-oriented booking dialogue but cannot express open-ended free-text fields — a clinical diagnosis or a doctor's recommendation cannot be reduced to a small enumerable set of slot values without losing information.

How this project differs. FnCTOD's JSON function-call framing is the direct ancestor of this project's JSON output condition in RQ3. The project's schema is deliberately hybrid: three list-valued fields (symptoms, family_history, medications) behave like FnCTOD's closed slots, while two fields (diagnosis, recommendation) are left as open free text — bridging FnCTOD's closed slot-value schema and ODIE's fully open table extraction.

2.4 InstructUIE — Wang et al., arXiv 2023 [3]

Approach. Wang et al. unify named-entity recognition, relation extraction, and event extraction under a single instruction-tuned FlanT5-11B model, jointly fine-tuned across 32 datasets. InstructUIE achieves 85.19% average NER F1 against an 80.09% BERT baseline, outperforms GPT-3.5 zero-shot on relation extraction, and generalises to seven NER domains never seen during its own fine-tuning.

Merit. InstructUIE's seven-domain zero-shot generalisation experiment is the direct precedent for this project's seen/unseen specialty split and RQ4 generalisation analysis, demonstrating that instruction-tuned models can be evaluated meaningfully on generalisation to excluded categories, not merely in-distribution performance.

Weakness and gap. InstructUIE reports strong aggregate accuracy but no hallucination metric and no comparison against a function-calling output representation — its own output format is natural-language, label-based text. Nor does it address dialogue.

How this project differs. InstructUIE's natural-language label output is the direct ancestor of this project's “NL” output condition in RQ3. This project's finding that the NL-label format hallucinates substantially more (33.8–58.0%) than a JSON function-calling alternative (7.5–9.7%) is a direct, empirical extension of the evaluation gap InstructUIE itself leaves open.

2.5 GoLLIE — Sainz et al., ICLR 2024 [4]

Approach. Sainz et al. argue that LLMs should be trained to follow explicit annotation guidelines, not just a bare task instruction. Guidelines are represented as Python dataclasses, with each field's docstring carrying its definition and a commented worked example. Five regularisation techniques — class-order shuffling, class dropout, guideline paraphrasing, candidate-label sampling, and class-name masking — stop the model from memorising label names as shortcuts. GoLLIE 7B outperforms an 11B InstructUIE by 13 F1 points on zero-shot benchmarks and hallucinates in under 1% of its predictions.

Merit. GoLLIE is the source of both the dataclass output format compared in RQ3 and the entire premise of RQ2. Its near-1% hallucination rate is direct evidence that hallucination is not solely a function of model size and can be engineered down.

Weakness and gap. GoLLIE is evaluated only on static news and biomedical text, never on dialogue, so its central claim — that guideline-conditioning improves generalisation — had never been tested against conversational, informally phrased input before this project's RQ2. Its 13-point improvement also uses the full five-technique regularisation regime at 7B–34B scale on high-end hardware, a materially larger investment than a single-consumer-GPU MSc project can reproduce.

How this project differs. RQ2 is a direct, deliberately simplified replication of GoLLIE's central claim in a setting GoLLIE never tested: conversational clinical dialogue, using a single technique (guideline text embedded in the fine-tuning prompt) rather than GoLLIE's full regularisation regime. The resulting +2.3-point gain is treated not as a failure to replicate, but as evidence — discussed in Chapter 6 — that GoLLIE's larger reported gain likely reflects its additional regularisation machinery rather than guideline text alone, a hypothesis this project's simpler ablation is well positioned to have surfaced precisely because it isolates guideline presence as the sole variable.

2.6 Complementary Technical and Corpus Literature

Beyond the four core methodological papers reviewed above, six further sources supply the technical and corpus foundations this project builds on directly.

Parameter-efficient fine-tuning. LoRA [7] freezes a pretrained model’s weights entirely and injects small, trainable low-rank decomposition matrices into selected linear layers, reducing the number of trainable parameters by orders of magnitude relative to full fine-tuning while retaining most of its benefit — the technique underlying every fine-tuning run in this project (Chapter 4). QLoRA [8] extends this further with 4-bit NF4 quantisation of the frozen base weights, together with double quantisation and paged optimisers to control memory spikes during training. This is the specific combination of techniques that made fine-tuning an 8-billion-parameter model feasible at all on a single 8 GB consumer laptop GPU: without QLoRA’s quantisation scheme, Qwen3-8B alone would not fit in the available VRAM budget, let alone leave room for gradients and optimiser state.

Backbone model. The Qwen3 technical report [9] documents the architecture, training regime, and reported benchmark performance of the model family used throughout this project, for both the 8B backbone and the smaller 1.7B variant used in the supplementary model-size comparison (Section 6.8). Its reported strong instruction-following at the 7–8B parameter scale is the direct justification for selecting Qwen3-8B over a larger, less locally deployable alternative.

Source corpus. MTS-Dialog [10] is the project’s source corpus: a publicly released collection of doctor-patient dialogue transcripts paired with human-written clinical summaries, built originally for clinical note-generation research. This project repurposes those human-written summaries in a role their original authors did not intend — as an independent cross-check during the self-verification quality-filtering pass (Chapter 4) — extracting additional evidentiary value from an existing, already-curated resource rather than requiring separately collected ground truth.

Task-oriented dialogue paradigm. Budzianowski and Vulic [5] and the MultiWOZ 2.2 paper [6] together establish the neural, pretrained-language-model paradigm for task-oriented dialogue systems and the benchmark that paradigm is most commonly evaluated against. They are not sources this project’s methodology is built on directly, but they supply the field-level context motivating its opening claim: that zero-shot dialogue-state tracking trails fine-tuned models by a substantial, well-documented margin on MultiWOZ, framing how large a fine-tuning-versus-prompting gap is typically expected to matter in dialogue tasks generally. They also represent the kind of standard, multi-domain dialogue benchmark this project’s original plan intended to incorporate before the scope was narrowed to a single clinical domain (Chapter 7).

2.7 Comparative Literature Matrix

Table 1. Comparative matrix of the four core reviewed papers by theme, evidence unit, and the limitation each carries into this project’s own design.

Paper	What it predicts / controls	Evidence unit	Limitation carried into this project
ODIE [1]	User-instructed open-schema table extraction	Static free text (news/web)	No multi-turn context; no negation/attribution handling needed on static text
FnCTOD [2]	Closed-schema dialogue-state as a function call	Task-oriented dialogue	Cannot express open-ended free-text fields (diagnosis, recommendation)

InstructUIE [3]	Unified NER/relation/event extraction	Static text, 32 datasets	No hallucination metric; NL-label output not compared against a structured alternative
GoLLIE [4]	Guideline-conditioned zero-shot extraction	Static news/biomedical text	Never tested on dialogue; full 5-technique regularisation not reproducible on this project's hardware budget

Source: Compiled from the four papers' own reported methods and evaluation sections.

2.8 Synthesis: Positioning and Novelty of the Present Work

Read together, the four core papers reveal a research space explored along individual axes but never combined. ODIE and InstructUIE demonstrate that structured extraction and cross-domain generalisation are achievable with fine-tuned open models, but neither addresses dialogue. FnCTOD addresses dialogue directly and popularises function-calling as an output representation, but only within closed, fixed slot schemas. GoLLIE addresses guideline-conditioning and output-format hallucination control with the strongest results of the four, but only on static text. No existing work in this review combines fine-tuning-versus-prompting comparison, guideline conditioning, and output-format hallucination control within a single conversational, open-ended-field benchmark evaluated under identical conditions. This project's contribution is precisely that combination, instantiated in a single high-stakes domain, with RQ1 (informed by ODIE and FnCTOD), RQ2 (informed by GoLLIE), and RQ3 (informed by all three output-format papers) tested on identical held-out data so their effects can be compared directly against one another — something none of the four source papers' own, mutually non-comparable benchmarks permits.

Chapter 3 — Problem Statement and Mathematical Formulation

This chapter defines the extraction task, the three structured output representations compared in RQ3, and every scoring metric used throughout this report and the Research Paper, independent of any particular model or experimental run. All definitions below are taken directly from the project's own scoring implementation (`common_extraction.py`), so the equations reported here are the equations the reported numbers were actually computed with, not a post-hoc restatement.

3.1 Task Definition

Let c denote a conversation: an ordered sequence of turns $c = (u_1, u_2, \dots, u_n)$, where each turn u_i is attributed to either the doctor or the patient speaker role. Each conversation in the benchmark carries a gold annotation g , a structured record over five fields: symptoms (a list of $\{\text{name}, \text{duration}, \text{severity}\}$ objects), family_history (a list of $\{\text{relation}, \text{condition}\}$ objects), medications (a list of $\{\text{name}, \text{dose}, \text{frequency}\}$ objects), diagnosis (free text), and recommendation (free text). A model under evaluation, given c and a format-specific prompt, produces a predicted record p over the same five fields. The extraction task is to produce p such that every fact in p is (i) present in c , (ii) correctly attributed to the patient rather than a third party, and (iii) not a fact the patient explicitly denied.

3.2 Structured Output Representations

Three surface representations of the same underlying record p are compared in RQ3, each with its own parser in `common_extraction.py`:

JSON function-calling: p is emitted as a JSON object matching the five-field schema exactly, parsed by regex-extracting the first balanced $\{\dots\}$ block from the model's raw output and validating it with `json.loads`.

Natural-language (NL) labels: p is emitted as delimiter-separated labelled lines (e.g. “symptoms: headache (duration: two days, severity: severe); ...”), parsed with a purpose-built delimiter parser rather than a general-purpose format.

Python dataclass: p is emitted as a literal `MedicalRecord(...)` constructor call, parsed with a safe, non-evaluating AST visitor that walks the parsed syntax tree and reads back field values without executing arbitrary code from model output.

Structural validity for a given output o is a binary indicator: $\text{valid}(o) = 1$ if the format-specific parser returns a well-formed record, 0 (a parse failure) otherwise. Structural-validity rate over a set of N outputs is the mean of $\text{valid}(o)$ across that set.

3.3 Fuzzy Slot-Level F1 for List Fields

For each list-valued field (symptoms, family_history, medications), predicted items and gold items are matched using token-Jaccard overlap rather than exact string equality, so a paraphrase (“kids” vs. “children”) still counts as a match. For two strings a and b , let A and B be their normalised (lower-cased, punctuation-stripped) token sets. The fuzzy-match predicate is:

$$\text{fuzzy_match}(a, b) = 1 \text{ if } |A \cap B| / |A \cup B| \geq 0.5, \text{ else } 0. \quad (1)$$

Matching proceeds greedily: each predicted item's key string (e.g. a symptom's name; a family-history item's “relation condition” concatenation) is compared in order against all not-yet-matched gold items, and the first fuzzy match found is consumed. Given true positives tp (matched pairs), $\text{precision} = tp / |\text{predicted}|$, $\text{recall} = tp / |\text{gold}|$, and the field's F1 is the harmonic mean:

$$F1_field = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall}). \quad (2)$$

3.4 Token-Overlap F1 for Free-Text Fields

The two free-text fields (diagnosis, recommendation) are scored with standard precision/recall over normalised token sets, rather than the greedy item-matching used for list fields, since each is a single string rather than a list of discrete facts: precision is the fraction of predicted tokens that also appear in the gold text, recall is the fraction of gold tokens that also appear in the predicted text, and F1 is again their harmonic mean (Equation 2). By convention, two empty strings score $F1 = 1.0$ (correctly predicting “nothing stated”), while one empty and one non-empty string score $F1 = 0.0$.

3.5 Macro-F1

The headline comparison metric used throughout this project is macro-F1: the unweighted arithmetic mean of the five per-field F1 scores for a given item, then averaged again across all items in the evaluated set:

$$macro-F1 = (F1_symptoms + F1_family + F1_meds + F1_diagnosis + F1_recommendation) / 5. \quad (3)$$

Macro-F1 (rather than a token-count-weighted average) was chosen specifically so that a field with typically shorter content, such as medications, is not swamped in the aggregate score by a field with typically longer content, such as symptoms — each of the five clinical fact types is treated as equally important to get right.

3.6 Hallucination Rate

Hallucination rate is defined as the fraction of predicted facts whose normalised tokens do not appear anywhere in the source conversation's normalised token set. For a prediction p with extracted facts $F = \{f_1, \dots, f_k\}$ (every symptom name, family-history condition, medication name, plus the diagnosis and recommendation strings, when non-empty), and conversation token set C :

$$hallucination_rate(p, c) = |\{f \in F : tokens(f) \cap C = \emptyset\}| / |F|. \quad (4)$$

This is a grounding proxy, not a clinical-correctness check: a fact can be grounded (its tokens appear in the conversation) while still being wrongly attributed or negated, which is why hallucination rate is reported alongside, never instead of, macro-F1 throughout this project.

3.7 Constraints and Evaluation Objective

The evaluation objective is not simply to maximise macro-F1. It is to select an adaptation strategy — prompting, fine-tuning, or fine-tuning with guidelines — and an output format whose macro-F1, hallucination rate, and structural validity are jointly acceptable, and to characterise, rather than hide, how that joint acceptability changes between specialties seen during any fine-tuning and specialties held out entirely (RQ4). The mathematical formulation therefore keeps macro-F1, hallucination rate, and structural validity as three separate reported quantities rather than collapsing them into one scalar, so that a format or condition which trades one for another (as RQ3's few-shot interaction with hallucination shows, Chapter 6) remains visible rather than averaged away.

Chapter 4 — Proposed Methodology and System Design

4.1 End-to-End Pipeline Architecture

The project is organised as one linear pipeline with a fork at the quality-filtering stage. Figure 2 shows the full path: source corpus, LLM annotation, the two competing filtering strategies (one abandoned,

one adopted), human review, the locked benchmark, and the four parallel experimental tracks that consume it.

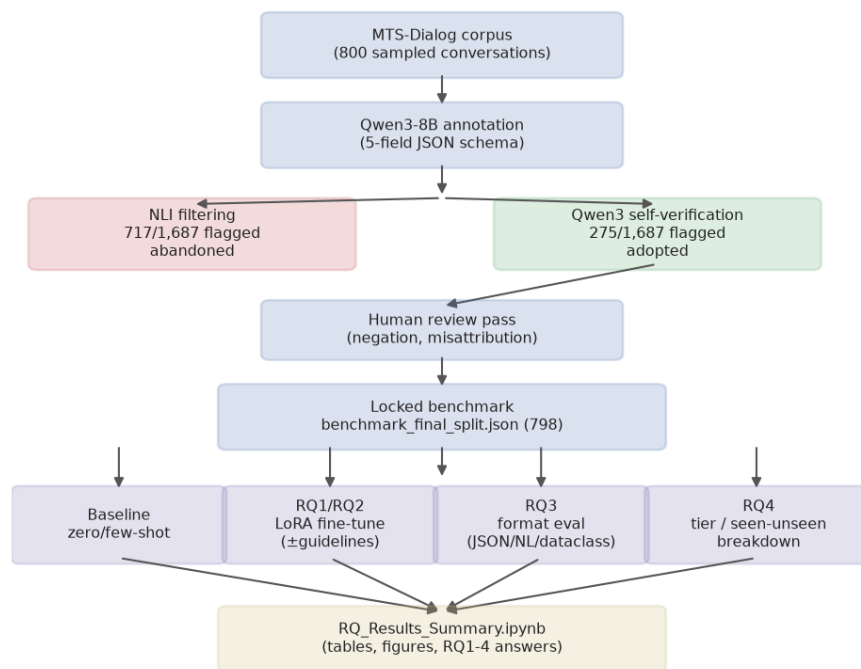


Figure 2. End-to-end pipeline architecture, from source corpus through benchmark construction to the four parallel experimental tracks (baseline, RQ1/RQ2 fine-tuning, RQ3 format evaluation, RQ4 stratified analysis) and results aggregation.

Source: Report-only diagram, derived from the project's own notebook and script sequence (*Annotation_Pipeline_Qwen3.ipynb* through *RQ_Results_Summary.ipynb*).

4.2 Annotation Prompt Design

Every benchmark item begins as a raw conversation passed to Qwen3-8B with an extraction prompt engineered specifically to handle two recurring error types identified during early trial runs. The prompt requires the model to attend to who each fact belongs to — the symptoms field is reserved strictly for the patient's own current experience, never a family member's — and to negation: a symptom the doctor asks about and the patient denies must not appear anywhere in the output. The operative rules, taken directly from the deployed prompt, read:

“The 'symptoms' field is ONLY for things the patient THEMSELVES is currently experiencing... Do not put a family member's condition, history, or behavior in 'symptoms' — even if it sounds like a symptom... The doctor often asks about a symptom and the patient DENIES it... A denied/negated symptom means the patient does NOT have it — do NOT include it.”

This exhibit is quoted directly from ANNOTATION_PROMPT in the project's own annotation notebook, reproduced here as evidence that the negation- and attribution-handling described throughout this report is a real, deployed constraint on the annotator, not a post-hoc description of desired behaviour.

4.3 Guideline-Conditioned Prompt Design for RQ2

RQ2 tests whether embedding explicit, per-field annotation guidelines in the fine-tuning prompt — rather than only a bare task instruction — improves generalisation to unseen specialties, replicating GoLLIE's central claim [4] in a conversational setting GoLLIE itself was never tested on. The guideline text actually used, taken directly from GUIDELINES_PROMPT in common_extraction.py, defines each field as follows:

Table 2. Per-field annotation guidelines embedded in the RQ2 fine-tuning prompt.

Field	Guideline definition (verbatim)
symptom	Patient-reported complaint (not one they deny). name; duration ("" if unstated); severity ("" if unstated).
family_history	A relative's condition, explicitly mentioned only — not the patient's own. relation (e.g. "father"); condition.
medication	A drug currently taken or prescribed (not one stopped/avoided). name; dose ("" if unstated); frequency ("" if unstated).
diagnosis	Clinician's stated diagnosis, free text, "" if none stated.
recommendation	Clinician's stated plan/advice, free text, "" if none given.

Source: common_extraction.py, GUIDELINES_PROMPT (verbatim field definitions).

This is the entire independent variable between the RQ1 (no-guidelines) and RQ2 (with-guidelines) fine-tuning runs: identical 250-example training subsample (fixed seed = 42), identical LoRA configuration, identical training arguments — only these per-field definitions are added to the prompt. Isolating guideline presence as the sole variable is what allows RQ2's +0.023 macro-F1 / −1.5-point hallucination result to be attributed to the guideline text itself, rather than confounded with any other change.

4.4 Quality-Filtering Architecture

Two automated filtering strategies were implemented and trialled before human review, as shown in Figure 2. The first, an NLI-based consistency check, converted each extracted fact into a natural-language claim (e.g. “The patient currently has {symptom}.”) and tested it against the conversation using cross-encoder/nli-deberta-v3-base, flagging a fact when the classifier returned “contradiction” above 0.5 confidence or “neutral” above 0.7 confidence. Run across all 800 items, this flagged 717 of 1,687 facts (42.5%) — on manual inspection, a substantial share of these were in fact correct, indicating the classifier, trained on single-sentence premise/hypothesis pairs, generalised poorly to long, multi-turn medical dialogue where a fact stated early can be clarified or confirmed many turns later.

The second strategy, adopted in its place, uses Qwen3-8B itself as a verifier: for each item, every extracted fact is listed and the model is asked to mark each TRUE or FALSE strictly against the conversation, with explicit instructions to mark FALSE any fact that is unmentioned, explicitly denied, or misattributed to the wrong person. This flagged 275 of 1,687 facts (16.3%) — a more plausible rate on the same manual-inspection standard — and every flagged fact was then reviewed by hand, corrected where necessary (most commonly negation errors and one misattribution case), and merged back into the final benchmark.

4.5 Model Architecture and LoRA Configuration

The backbone model is Qwen3-8B [9], an open-source instruction-tuned decoder-only transformer, selected for strong reported instruction-following at the 7–8B scale, open weights permitting full gradient access unavailable with a proprietary API, and mature LoRA/PEFT tooling. The model is loaded in 4-bit precision (NF4 quantisation with double quantisation, float16 compute) [8], the specific choice that made loading and fine-tuning an 8-billion-parameter model feasible within an 8 GB VRAM budget. A LoRA adapter [7] is attached to the four attention projection matrices (query, key, value, output) with rank 16, scaling alpha 32, and dropout 0.05, training 15,335,424 of 8,206,070,784 parameters (0.187%) while leaving base weights frozen. Rank, alpha, and target-module selection follow established LoRA and GoLLIE defaults [4], [7] rather than a project-specific search, a trade-off discussed critically in Chapter 8.

4.6 Training Procedure

Training data for the RQ1 (no-guidelines) condition comprises 250 of the 709 seen examples, drawn with a fixed random seed (42) for reproducibility. Each training example pairs the extraction prompt with the gold JSON annotation as the target completion; prompt tokens are masked to label value -100 during tokenisation, so cross-entropy loss is computed exclusively over the target JSON completion, not the input. Training uses a per-device batch size of 1 with gradient accumulation over 8 steps (effective batch size 8), a single epoch, learning rate $2e-4$, fp16 mixed precision, the paged 8-bit Adam optimiser, and gradient checkpointing enabled throughout. For RQ2, an otherwise identical script substitutes the guideline-augmented prompt (Section 4.3) for the plain extraction prompt, holding every other hyperparameter fixed.

Chapter 5 — Dataset, Implementation and Experimental Design

5.1 Accepted Dataset Facts

The benchmark was constructed from 800 conversations sampled from the MTS-Dialog corpus [10], a publicly available collection of doctor-patient dialogue transcripts developed for clinical note-generation research. After the annotation, filtering, and human-review stages described in Chapter 4, the locked benchmark (benchmark_final_split.json) comprises 798 examples — two of the original 800 were dropped during review — stratified along three independent dimensions.

Table 3. Composition of the locked benchmark (798 examples).

Dimension	Breakdown
Total examples	798
Specialty split	Seen: 709 (88.8%) Unseen: 89 (11.2%)
Difficulty tier	Easy: 299 Medium: 257 Hard: 242
Held-out specialties	Dermatology, Orthopedics (excluded from all fine-tuning)

Source: benchmark_final_split.json; RQ_Results_Summary.ipynb aggregate counts.

Table 4. Sub-specialty distribution across the 798-example benchmark.

Sub-specialty	n	Share
General medicine	266	33.3%
Cardiology	104	13.0%
ENT	94	11.8%
Gastroenterology	60	7.5%
Orthopedics (unseen)	59	7.4%
Neurology	51	6.4%
Endocrinology	35	4.4%
Dermatology (unseen)	30	3.8%
Pulmonology	26	3.3%
Mental health	22	2.8%
Gynecology	18	2.3%
Oncology	17	2.1%
Urology	16	2.0%

Source: Benchmark construction notebook aggregate counts (rough project review draft).

The 88.8/11.2 seen/unseen split and the two held-out specialties were fixed before any fine-tuning experiment and reused, unchanged, across RQ1, RQ2, and RQ4, so generalisation results are directly comparable across research questions. The imbalance visible in the sub-specialty table — general medicine at 33.3% against oncology and urology at roughly 2% each — is revisited as a fairness concern in Chapter 8.

5.2 Preprocessing and Seeds

Conversations are parsed from raw “Doctor:” / “Patient:” prefixed transcript text into a turn-structured list before being passed to any prompt, so every downstream stage (annotation, verification, evaluation) consumes the same turn representation. Random seed 42 is used consistently for the RQ1/RQ2 250-example training subsample, so the identical subsample underlies both fine-tuning conditions (Chapter 4). No additional resampling or held-out reshuffling was performed once the benchmark’s seen/unseen split was locked, to avoid any risk of test-set information leaking into a later model-selection decision.

5.3 Implementation Environment

Table 5. Hardware and software configuration for all reported experiments.

Component	Specification
GPU	NVIDIA RTX 5060 Laptop GPU (8 GB VRAM)
OS	Windows 11
PyTorch	2.11.0+cu128 (Blackwell sm_120 kernel support)
Transformers	5.15.0
PEFT	0.20.0
bitsandbytes	0.50.0
Base model	Qwen/Qwen3-8B and Qwen/Qwen3-1.7B (Hugging Face)

Source: Verified via an explicit GPU matrix-multiplication smoke test, not version metadata alone (Section 5.5).

All experiments were run on this single local machine, following an initial migration away from free-tier Google Colab, whose shared GPU allocation proved unreliable for multi-hour fine-tuning runs. Reported timings: RQ1 fine-tuning took approximately 42 minutes of training plus 39 minutes of inference; RQ2 fine-tuning, after the fixes described in Section 5.5, took 108.9 minutes of training at ~100% GPU utilisation (~7.2 GB VRAM) plus 63.4 minutes of inference on the 89-item unseen split.

5.4 Experimental Protocol

Table 6. Experimental protocol and evidence role per research question.

RQ	Conditions compared	Evaluated on	Independent variable
Baseline	Zero-shot; few-shot (1 example)	All 798 items	Presence of a worked example in the prompt
RQ1	Zero-shot; few-shot; LoRA fine-tuned (no guidelines)	89-item unseen split	Prompting vs. fine-tuning

RQ2	Fine-tuned, no guidelines; fine-tuned, with guidelines	89-item unseen split	Presence of per-field guideline text (Section 4.3)
RQ3	JSON; NL labels; Python dataclass, each zero-/few- shot	All 798 items	Output representation
RQ4	Zero-shot, grouped by difficulty tier and specialty split	All 798 items	Conversational difficulty; specialty familiarity
Supplementary	Qwen3-1.7B vs. Qwen3- 8B, zero-/few-shot	All 798 items	Model parameter count

Source: *Baseline_Evaluation_Qwen3.ipynb*; *LoRA_Finetune_Qwen3.ipynb*; *run_format_eval.py*; *RQ_Results_Summary.ipynb*.

5.5 Threats to Validity, Stated Before Results

Two classes of threat to validity are stated here, before any result is presented in Chapter 6, rather than discovered retrospectively by a reader comparing tables.

Infrastructure threats

`torch.cuda.is_available()` returning True is not sufficient evidence that GPU computation will succeed: one candidate environment reported CUDA availability while every real matrix multiplication crashed, because the installed PyTorch build lacked compiled kernels for the RTX 5060's Blackwell (`sm_120`) architecture. A second, separately maintained conda environment was found to hold a CPU-only PyTorch build with none of transformers, peft, bitsandbytes, or datasets installed, despite being recorded elsewhere as “working.” Both were caught by an explicit GPU matrix-multiplication smoke test before any experiment in this report was trusted to have actually run on GPU.

Data and protocol threats

A tokenisation-length threat was diagnosed during the RQ2 training run: the longer guideline-augmented prompts caused 177 of 250 training examples (71%) to exceed the original 640-token cap, silently truncating — and in some cases entirely discarding — the target JSON completion. This was caught by auditing the tokenized length distribution directly and fixed by compacting the guideline text and raising the cap to 768 tokens, restoring the truncation rate to 8% (20/250), matching the original RQ1 run's rate on the same metric. A second threat concerns sample size: the RQ1/RQ2 headline comparisons rest on only 89 unseen items, and no formal significance test has yet been run on the observed deltas (Chapter 8); a 1–2-point macro-F1 difference on this sample should be read as suggestive, not conclusive, until that validation is performed. A third threat concerns correlated bias: the initial annotation and the self-verification quality-filtering pass (Chapter 4) both use the same Qwen3 model family, so a systematic bias in how that model perceives a given specialty's terminology could, in principle, be encoded into the gold labels themselves rather than caught by a genuinely independent check. The subsequent human review pass mitigates, but does not eliminate, this risk.

Chapter 6 — Results and Critical Discussion

6.1 Baseline: Zero-Shot versus Few-Shot

Table 7. Baseline Qwen3-8B zero-shot / few-shot macro-F1, all 798 items.

Field	Zero-shot F1	Few-shot F1
Symptoms	0.8171	0.8091
Family history	0.8339	0.8317
Medications	0.9632	0.9571
Diagnosis	0.7429	0.8560
Recommendation	0.7563	0.8165
Macro average	0.8227	0.8541

Source: *evaluation_results_zeroshot.csv*; *evaluation_results_fewshot.csv*. Parse failures: 3/798 (0.4%) zero-shot, 1/798 (0.1%) few-shot.

This is the starting point every other condition in this chapter is compared against: can the base model perform the extraction task at all, with no fine-tuning? The answer is yes, reasonably well — 0.823 macro-F1 zero-shot, rising to 0.854 with a single worked example. Medications is the easiest field (0.96+ in both modes, likely because drug names are lexically distinctive and rarely ambiguous); diagnosis and recommendation are the fields that benefit most from a worked example (+0.11 and +0.06 F1 respectively), consistent with these being the two free-text fields where a model has the most latitude to phrase things differently from the gold label.

6.2 Worked Example 1: Negation Handling and Its Limits

The following case study, drawn directly from the annotation pipeline's own recorded output for benchmark item ID 34, illustrates both the negation-handling rule working as intended and a residual limitation it did not fully resolve.

Conversation (verbatim):

DOCTOR: Have you lost weight? / PATIENT: No, I haven't, you can ask my daughter. / DOCTOR: How about anything like fever, nausea, vomiting, pain? / PATIENT: No. / DOCTOR: And as we discussed you have problem talking and comprehending. / PATIENT: Yes. / DOCTOR: And you get urine infections too? / PATIENT: Yes. / DOCTOR: And as discussed, your drainage is similar to past episodes when you had such infection? / PATIENT: Yes.

Recorded annotation output:

symptoms: [{"name": "difficulty talking and comprehending", "duration": "similar to past episodes", "severity": "not specified"}]; family_history: [{"relation": "daughter", "condition": "weight loss"}]; medications: []; diagnosis: ""; recommendation: "".

The negation rule worked correctly in one respect: “weight loss” — explicitly denied by the patient (“No, I haven’t”) — was correctly excluded from the symptoms field, where a naive extractor would otherwise have recorded it simply because the word appeared in the doctor's question. However, the annotation still shows a residual artefact: the denied weight loss was folded into family_history, attributed to “daughter,” apparently because the patient's aside (“you can ask my daughter”) was misread as introducing a fact about the daughter rather than merely offering a witness to confirm the patient's own denial. The annotation also misses two facts the patient explicitly confirmed — urine infections and drainage similar to past episodes — which do not appear anywhere in the recorded output. This example is included here, rather than a cleaner one, specifically because the project's own critical-

appraisal standard (Chapter 8) is to report evidence of residual error alongside evidence of success, rather than present only favourable cases.

6.3 Worked Example 2: Self-Verification Catching a Temporal Attribution Error

This second case study, drawn from the self-verification pass's own recorded output for benchmark item ID 566, illustrates the specific class of error self-verification is designed to catch — one that negation-only rules cannot.

Conversation excerpt (verbatim):

DOCTOR: When was your last stroke? / PATIENT: Well, it was on January seventeenth. I had these episodes where I was like always so confused and looks like I lost everything. All my memory and everything. And then they did a scan. It was verified that I was having a stroke on the same day. / DOCTOR: Did they put you on some medications? / PATIENT: They gave me an anticonvulsant and aspirin. / [...] / DOCTOR: Do you feel anything else along with these episodes... any kind of weakness or numbness or shortness of breath? / PATIENT: No nothing else, just the confusion and forgetfulness.

Facts sent for self-verification, and Qwen3-8B's verdicts:

Table 8. Self-verification facts and verdicts for benchmark item ID 566.

#	Fact sent for verification	Verdict
1	The patient currently has: confusion	TRUE
2	The patient currently has: forgetfulness	TRUE
3	The patient currently has: episodic confusion and memory loss	TRUE
4	The patient takes/took: aspirin	TRUE
5	The patient takes/took: dyphenhydramine	TRUE
6	The patient's diagnosis is: stroke	FALSE

Source: *Qwen_self_verification.ipynb*, recorded raw verification response for item ID 566.

The verifier flagged “stroke” as FALSE for the diagnosis field even though the patient does state a stroke was “verified” by a scan. The distinction the verifier appears to be drawing — correctly, on inspection — is temporal: the confirmed stroke belongs to a past episode (“last stroke... on January seventeenth”) being recounted as history, not a diagnosis made during the conversation being annotated, whose actual subject is the patient's recurring confusion/forgetfulness episodes. This is exactly the class of error a simple negation check cannot catch, because nothing in the sentence is denied — the problem is temporal and contextual attribution, not polarity. It is offered here as concrete evidence for why the self-verification approach, despite being a coarser tool than the abandoned NLI classifier on paper, outperformed it in practice: it can reason about context an entailment classifier trained on isolated sentence pairs cannot.

6.4 RQ1 Results: Fine-Tuning versus Zero-/Few-Shot Prompting

Table 9. RQ1: zero-shot vs. few-shot vs. LoRA fine-tuned, unseen split (n = 89).

Condition	n	Macro-F1
Zero-shot	89	0.756
Few-shot	89	0.776
Fine-tuned (LoRA, no guidelines)	89	0.757

Source: `finetune_comparison_summary.csv`.

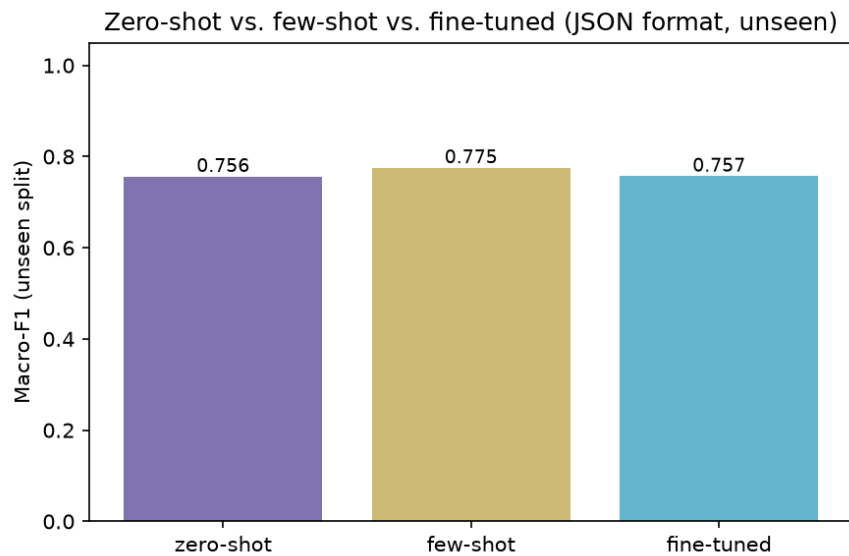


Figure 3. Zero-shot vs. few-shot vs. fine-tuned macro-F1 on the unseen split (RQ1).

Source: `figures/finetune_comparison.png`, generated by `RQ_Results_Summary.ipynb`.

Fine-tuning essentially matched zero-shot prompting (+0.001 F1) and underperformed few-shot prompting (−0.019 F1) on the unseen split — a genuine null result relative to the project plan's anticipated 5–15-point improvement, informed by ODIE's ~5.4% gain and FnCTOD's pattern of fine-tuned models matching ChatGPT. It is reported here as a null result rather than reframed, because the practical conclusion it supports — that a small example budget may be better spent as few-shot context than as fine-tuning data — is itself a concrete, actionable finding, discussed further in Section 6.10 and Chapter 8.

6.5 RQ2 Results: Annotation-Guideline Ablation

Table 10. RQ2: guideline-conditioned vs. non-guideline fine-tuning, unseen split (n = 89).

Condition	n	Macro-F1	Hallucination
Zero-shot	89	0.756	—
Few-shot	89	0.776	—
Fine-tuned, no guidelines	89	0.757	12.8%
Fine-tuned, with guidelines	89	0.780	11.2%

Source: *rq2_guidelines_comparison_summary.csv*.

Adding the per-field guideline text shown in Section 4.3 to the fine-tuning prompt improved macro-F1 by +0.023 and reduced hallucination by 1.5 percentage points on the unseen split — the best-performing condition observed across RQ1 and RQ2 combined, though more modest than the 8–13-point improvement GoLLIE reports for its own, more heavily regularised regime [4]. Because the guideline text is the sole variable changed between this condition and the RQ1 fine-tuned condition (Section 4.3), this gain can be attributed specifically to guideline presence rather than to any confound in training data, LoRA configuration, or training arguments.

6.6 RQ3 Results: Output-Format Comparison

Table 11. RQ3: output-format comparison across all 798 items.

Format / mode	Struct. valid.	Halluc. rate	Macro-F1
JSON, zero-shot	99.6%	9.7%	0.823
JSON, few-shot	99.9%	7.5%	0.854
NL labels, zero-shot	99.7%	33.8%	0.592
NL labels, few-shot	100.0%	58.0%	0.429
Dataclass, zero-shot	98.2%	12.1%	0.747
Dataclass, few-shot	98.6%	26.1%	0.680

Source: *rq3_format_comparison_summary.csv*.

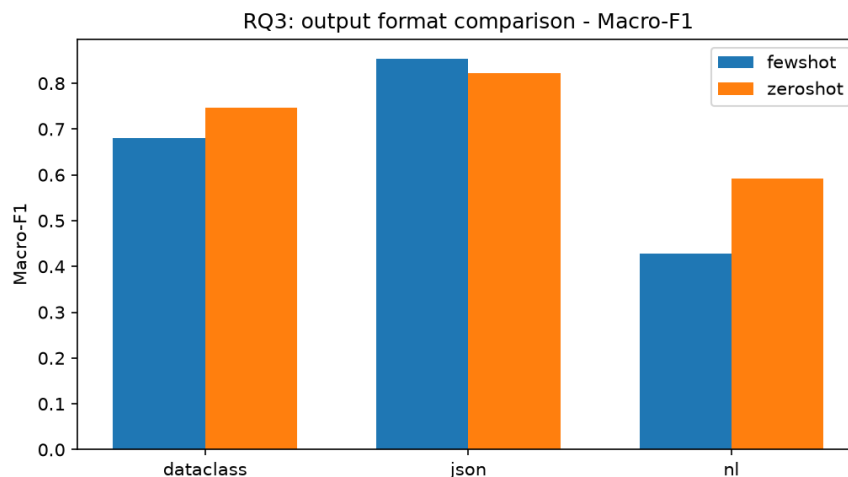


Figure 4. RQ3 macro-F1 by output format and prompting mode.

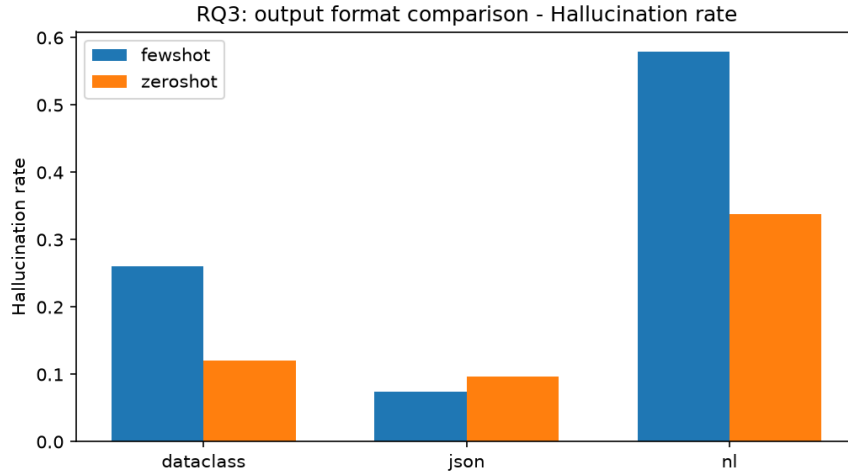


Figure 5. RQ3 hallucination rate by output format and prompting mode.

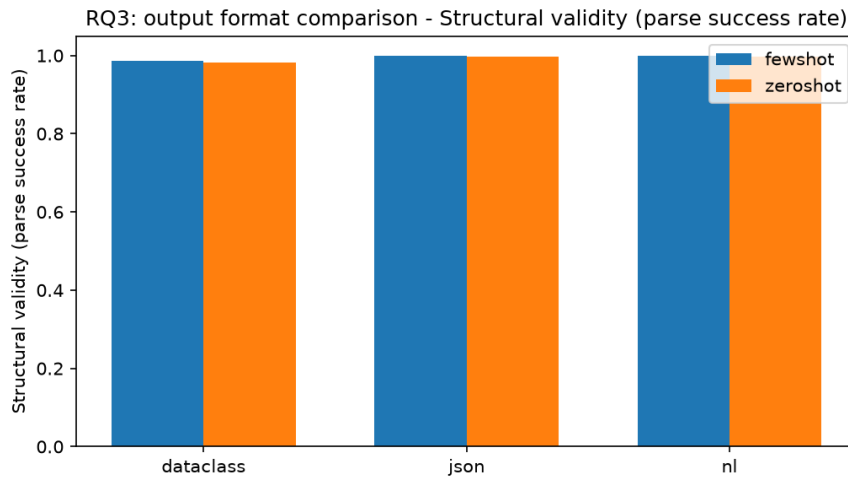


Figure 6. RQ3 structural validity (parse success rate) by output format and prompting mode.

JSON wins decisively on every metric — highest structural validity, lowest hallucination rate, and highest macro-F1 in both prompting modes — matching the project plan's anticipated result exactly. The most striking secondary finding is the divergent effect of a worked example on hallucination: adding one reduces JSON's hallucination rate (9.7% $\rightarrow$ 7.5%) but increases it sharply for NL labels (33.8% $\rightarrow$ 58.0%) and dataclass (12.1% $\rightarrow$ 26.1%). A plausible mechanism is that JSON's rigid, template-like grammar constrains what the example can teach the model to do — essentially only which facts to extract — whereas the freer generation spaces of NL labels and dataclass syntax let the example also teach a pattern of elaboration, which the model then applies even where the conversation does not support it. This interaction is not reported in any of the four core reviewed papers and is flagged as a concrete direction for follow-up mechanistic investigation.

6.7 RQ4 Results: Complexity-Tier and Generalisation Degradation

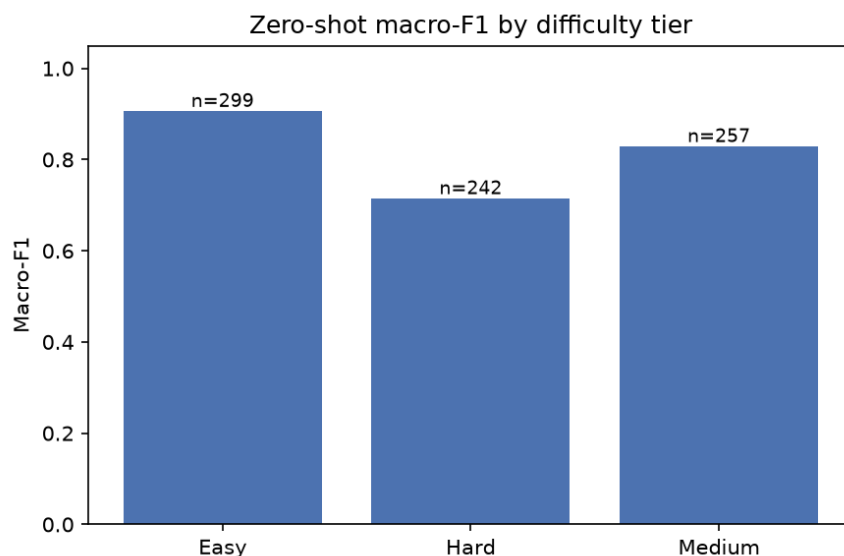


Figure 7. Zero-shot macro-F1 by conversational difficulty tier.

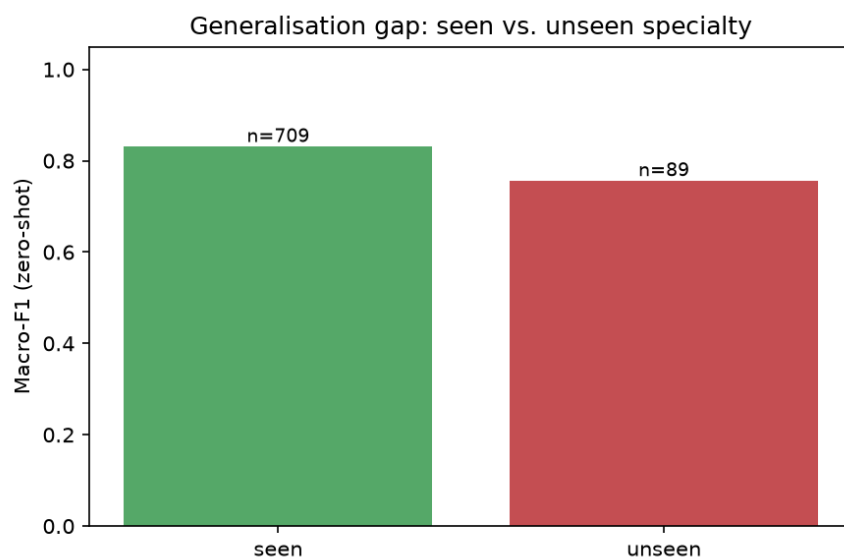


Figure 8. Generalisation gap: zero-shot macro-F1, seen vs. unseen specialty.

Zero-shot macro-F1 falls from 0.906 (Easy, $n = 299$) to 0.828 (Medium, $n = 257$) to 0.714 (Hard, $n = 242$) — a monotonic, 19-point decline across the three difficulty tiers, consistent with the intuitive expectation that longer, more ambiguous, or more multi-topic conversations are harder to extract from correctly. Separately, seen-specialty zero-shot macro-F1 (0.831) exceeds unseen-specialty macro-F1 (0.756) by 7.5 points. A manual review of the 30 lowest-scoring unseen examples classified 37% as scoring artefacts (ambiguous blank gold labels, singular/plural mismatches penalised by the exact-token matching underlying the fuzzy-F1 metric), 50% as genuine extraction misses, and 13% as mixed. The refined finding is that the true generalisation gap is smaller than the raw 7.5-point figure suggests, but

a real gap remains, concentrated in incomplete symptom lists and missed recommendations on the two unseen specialties.

6.8 Supplementary Result: Model-Size Comparison

Table 12. Qwen3-1.7B vs. Qwen3-8B, all 798 items (not requested by the original plan).

Model	Zero-shot F1	Few-shot F1
Qwen3-1.7B	0.772	0.773
Qwen3-8B	0.823	0.854

Source: *model_size_comparison_summary.csv*.

This comparison, added beyond the original plan's scope as a low-cost extension of the existing evaluation harness, quantifies the capability cost of a roughly 4.7x smaller model: a 5.1-point (zero-shot) to 8.1-point (few-shot) macro-F1 reduction. Notably, the smaller model gains almost nothing from a worked example (+0.001 F1) where the 8B model gains +0.031 — a plausible in-context-learning-capacity effect worth flagging for any future deployment cost/accuracy trade-off discussion, though not one this project's scope permits investigating further.

6.9 Comparison Against Anticipated Results

Table 13. Anticipated vs. actual results, RQ1–RQ3.

RQ	Anticipated	Actual
RQ1	+5 to +15 F1 pts over prompting	+0.001 pt (null result)
RQ2	+8 to +13 F1 pts	+2.3 pts (real, modest)
RQ3	JSON: lowest halluc., highest validity	Confirmed exactly

Source: *Project plan expectations vs. rq*_summary.csv results*.

RQ3's confirmation is unsurprising given its constrained, template-driven generation space. RQ1 and RQ2's shortfall against expectation is the more informative result, and is discussed critically in Chapter 8; the leading hypothesis is that 250 examples for a single epoch under-trains the adapter relative to the 700–7,200-example, multi-epoch regimes reported in ODIE, FnCTOD, and GoLLIE — a hypothesis this project's compute budget did not permit testing directly, but which is proposed as a concrete, falsifiable direction for future work (Chapter 9).

6.10 Integrated Answer to the Research Questions

Taken together, the evidence in this chapter answers each research question, but with boundaries that must be stated alongside the answer rather than after it. RQ1: naive LoRA fine-tuning on 250 examples does not outperform few-shot prompting on this benchmark — a negative result, not a null instrument, since the same pipeline detects a real, positive effect in RQ2 under an otherwise identical protocol. RQ2: embedding explicit annotation guidelines in the fine-tuning prompt produces a small but real, directionally consistent improvement in both accuracy and grounding, at effectively zero additional compute cost — the project's most actionable finding for a practitioner operating under a similarly constrained data and hardware budget. RQ3: JSON function-calling output should be the default

structured-output choice for a hallucination-sensitive extraction task, unambiguously, on every metric measured. RQ4: extraction quality degrades meaningfully with conversational difficulty, and a real, if partly artefactual, generalisation gap exists between clinical specialties represented during fine-tuning and specialties held out entirely — a gap this project's manual error review, rather than the raw aggregate metric alone, was needed to characterise honestly.

Chapter 7 — Project Lifecycle, Tools, Verification and Validation

7.1 Research Lifecycle as Decisions

The project began with a supervisor-approved project plan specifying an ambitious original scope: a conversational information-extraction benchmark spanning at least three domains (healthcare, customer support, e-commerce), comparison against proprietary baselines (GPT-3.5, GPT-4) and three literature baselines (InstructUIE, GoLLIE 7B, FnCTOD-LLaMA2-13B), and five research questions. Following a formal scope review with the supervisor, the plan's boundaries were explicitly narrowed: the domain was restricted to healthcare dialogue only, sourced from MTS-Dialog; the standard multi-domain comparison datasets (MultiWOZ 2.2, Schema-Guided Dialogue, IE INSTRUCTIONS) were dropped; no proprietary-model comparison was run, owing to the absence of an API budget; no literature-baseline reproduction was attempted, owing to the 8 GB GPU budget and the fact that two of the three literature baselines are themselves larger than Qwen3-8B; and a fifth research question (joint multi-format generalisation) was formally dropped, consistent with the plan's own risk-mitigation register, which permitted dropping RQ4/RQ5 once RQ1–RQ3 were judged solid.

Development then proceeded through the benchmark-construction stage described in Chapter 4: 800 conversations sampled and annotated, two competing filtering strategies trialled (one abandoned after inspection, the other adopted), a full human review pass, and a locked, stratified benchmark. Four parallel experimental tracks were then implemented against that locked benchmark — baseline evaluation, LoRA fine-tuning (RQ1, then RQ2), the three-format evaluation harness (RQ3), and the stratified RQ4 analysis — each producing a saved CSV of scores consumed unmodified by the final results-aggregation notebook. A hard submission deadline of 4 September 2026 was confirmed by the module supervisor, together with the requirement to prepare a Research Paper, this Supporting Report, a recorded presentation, and code as four separate final deliverables.

7.2 Tool Choices and Their Research Purpose

Table 14. Tools used to support development, grouped by purpose.

Tool	Purpose / relevance
NVIDIA RTX 5060 Laptop GPU (8 GB VRAM)	Local GPU for all fine-tuning and inference; its VRAM budget directly shaped the 4-bit quantisation, LoRA rank, and batch-size choices in Chapter 4.
Local Windows 11 machine	Replaced free-tier Google Colab after Colab's shared, unpredictable GPU allocation proved unreliable for multi-hour fine-tuning runs.
Qwen3-8B / Qwen3-1.7B (Hugging Face)	The primary backbone model and its smaller variant, used for annotation, self-verification, baseline evaluation, and every fine-tuning run.
cross-encoder/nli-deberta-v3-base	The first, ultimately abandoned, annotation-quality-filtering strategy (Chapter 4).
PyTorch 2.11.0+cu128	Core tensor/autograd framework; this exact build was required for compiled GPU kernels on the RTX 5060's Blackwell architecture.
Hugging Face transformers (5.15.0)	Model/tokenizer loading, chat-template application, and text generation for every Qwen3 call in the project.

Hugging Face peft (0.20.0)	LoRA adapter implementation, adapter save/load, and the final RQ2-adapter-to-base-model merge.
bitsandbytes (0.50.0)	4-bit NF4 quantisation with double quantisation, the technique that made an 8B-parameter model fit an 8 GB GPU.
Hugging Face datasets	Dataset formatting and batching for the fine-tuning loop.
pandas	CSV loading and aggregation throughout — every table in Chapter 6 is computed from a pandas groupby/mean over a saved evaluation_results_*.csv file.
Python standard library (json, re, ast)	json for benchmark/prediction I/O; re for regex JSON extraction; ast for a safe, non-evaluating dataclass parser.
common_extraction.py (custom module)	The project's own shared scoring module, imported unchanged by every scoring script so JSON, NL, and dataclass outputs are scored by identical logic against identical gold labels.
matplotlib	Every chart in figures/ and this report was produced with matplotlib, chosen for direct, scriptable control tracing every chart back to its source CSV.
openpyxl	Programmatic construction of the human-evaluation rating sheet, including a data-validation dropdown constraining ratings to the rubric's four valid levels.
scipy.stats	Pearson/Spearman/Kendall correlation, intended to validate the automatic fuzzy-F1 metric against human ratings once collected (Section 7.4).
Microsoft Word (conference-template-a4 template)	Authoring tool for the Research Paper, using the IEEE-style two-column conference template supplied for this module.
Gradio	Framework for the project's live interactive extraction demo (gradio_demo.py), used to visually verify RQ2 model behaviour against benchmark gold labels outside the aggregation notebook.

7.3 Verification and Provenance

Verification was treated as an explicit step at three separate points in the life cycle, rather than a single end-of-project check.

Data verification: raw LLM-generated annotations were verified twice, by two independent methods (NLI entailment, then LLM self-verification), before a full human review pass verified every fact either method flagged as uncertain — 275 flagged facts out of 1,687, rather than a full manual re-annotation of all 1,687.

Infrastructure verification: GPU computation was verified with an explicit matrix-multiplication smoke test rather than trusted from torch.cuda.is_available() alone, after this exact assumption was found

false in one environment (Chapter 5). Token-length truncation was verified by auditing the tokenized length distribution of the RQ2 training set directly.

Result verification: every headline figure reported in Chapter 6 and the Research Paper was re-derived directly from the project's saved CSV files by the results-aggregation notebook at the point of writing, rather than carried over by hand from an earlier draft.

Cross-condition validation: RQ1 and RQ2 share an identical 250-example training subsample (fixed seed = 42), identical LoRA configuration, and identical training arguments, differing only in whether the prompt includes guideline text — a controlled ablation design chosen specifically so RQ2's result could be attributed to guideline presence alone.

7.4 Reproducibility Lessons

A stratified 60-item human-evaluation package (rating sheet with a 4-level rubric adapted from ODIE, a separately kept answer key, and a correlation-scoring script) was built specifically to validate the automatic fuzzy-F1 metric against independent human judgement. This step was not completed at the time of writing and is reported here as an open limitation rather than omitted silently — a decision consistent with this report's wider policy of naming an incomplete or failed step explicitly rather than letting a tidy narrative erase it. The main reproducibility lesson carried forward is to make failure states first-class: the abandoned NLI-filtering run and its 717-fact flag list remain on disk rather than deleted, the RQ2 truncation bug's before/after token-length distributions remain available, and the incomplete human-evaluation package's built infrastructure (rating sheet, rubric, scoring script) is reported as built-but-unused rather than not mentioned.

7.5 Interactive Verification: The Live Extraction Demo

Beyond the results-aggregation notebook (Section 7.3), the project also provides a live, interactive verification surface: `gradio_demo.py`, a browser-based application built on the Gradio framework (Section 7.2, Table 14). The demo loads the RQ2 guideline-fine-tuned adapter after merging it into the Qwen3-8B base weights via `merge_lora_adapter.py`, and lets a reviewer select any of the 798 benchmark conversations by clinical section category, run extraction against it, and watch the model's structured output stream token-by-token as it is generated, rather than only as a finished block. On completion, the demo scores the output with `common_extraction.py` — the identical scoring module that produced every headline figure in Chapter 6 — computing per-field F1 against the conversation's gold annotation and its hallucination rate.

This closes the loop between the static tables reported above and a reproducible, inspectable instance of the same pipeline: a reviewer can verify, on demand, that the reported RQ2 metrics correspond to real model behaviour on genuine benchmark items rather than to the aggregation notebook's output alone. Figure 9 shows the interface after extracting a family-history conversation (benchmark item 9): the model's live-streamed output, the parsed JSON extraction, and the resulting per-field F1 and hallucination-rate scores against the benchmark's own gold answer. A screen recording of this demo in live operation forms part of the video presentation submitted alongside the Research Paper and this Supporting Report.

Clinical Information Extraction

Queen's University Belfast — MSc Themed Research Project

Paste a doctor/patient conversation (one SPEAKER: text: line per turn) and watch the model extract symptoms, family history, medications, diagnosis, and recommendation live.

Example conversations from the benchmark

Category

Pick a section type to list its conversations.

Fam / Sochx (234)

Conversation

Pick one to load it below (and get its gold reference + F1 score for free).

id 9 — 10 turns

Conversation

DOCTOR: Can you tell me about any diseases that run in your family?
PATIENT: Sure, my brother has a prostate cancer.
DOCTOR: Okay, brother.
PATIENT: My father had brain cancer.
DOCTOR: Okay, dad.
PATIENT: Then on both sides of my family there are many heart related issues.
DOCTOR: Okay.
PATIENT: And my brother and sister both have diabetes.
DOCTOR: Okay.
PATIENT: Yes, that's it.

Model output (live)

```
{
  "symptoms": [],
  "family_history": [
    {
      "relation": "brother",
      "condition": "prostate cancer"
    },
    {
      "relation": "father",
      "condition": "brain cancer"
    },
    {
      "relation": "brother",
      "condition": "diabetes"
    },
    {
      "relation": "sister",
      "condition": "diabetes"
    }
  ],
  "medications": [],
  "diagnosis": "",
  "recommendation": ""
}
```

(i) Parsed extraction

```
1 {
2   "symptoms": [
3   ],
4   "family_history": [
5     {
6       "relation": "brother",
7       "condition": "prostate cancer"
8     },
9     {
10      "relation": "father",
11      "condition": "brain cancer"
12    },
13    {
14      "relation": "brother",
15      "condition": "diabetes"
16    },
17    {
18      "relation": "sister",
19      "condition": "diabetes"
20    }
21  ],
22  "medications": [
23  ],
24  "diagnosis": "",
25  "recommendation": ""
26 }
```

Advanced: score your own conversation

Extract

Accuracy vs. ground truth

Hallucination rate works for any conversation. F1 accuracy needs a gold answer - either the benchmark's (automatic, for the examples below) or your own, typed above under 'Advanced'.

Gold reference / expected answer in use

His brothers had prostate cancer. Father had brain cancer. Heart disease in both sides of the family. Has diabetes in his brother and sister.

Field	F1
Symptoms	1.00
Family History	1.00
Medications	1.00
Diagnosis	1.00
Recommendation	1.00
Average	1.00

Hallucination rate: 50% (fraction of extracted facts not grounded in the conversation text)

Figure 9. The live extraction demo interface (gradio_demo.py), showing the category/conversation pickers, streamed model output, parsed JSON extraction, and per-field F1 / hallucination-rate scoring against the benchmark gold answer.

Chapter 8 — Critical Appraisal, Ethics and AI Disclosure

8.1 What Worked and What Underperformed

The strongest parts of this project are the separation of evidence layers, the explicit comparison between an abandoned and an adopted quality-filtering method (rather than presenting only the method that was kept), the decisive and literature-confirming RQ3 result, and the honest reporting of RQ1's null result. The 798-example benchmark is genuinely human-verified, not purely LLM-generated: every fact flagged by either filtering pass was reviewed and, where necessary, corrected by hand. The RQ2 adapter was also successfully fused into the base model's weights, producing a complete, standalone model that loads and runs inference directly with no separate PEFT adapter required at inference time — a working, runnable artefact of the project's fine-tuning work.

The underperforming part is fine-tuning itself, relative to what the project plan anticipated. RQ1's near-zero gain over zero-shot prompting, and its shortfall against few-shot prompting, means the compute and engineering investment in the base fine-tuning pipeline did not pay off in isolation — it took the

additional, low-cost guideline-conditioning intervention in RQ2 to produce a real, positive result. Scale-down in scope also cost the project its originally intended competitive framing: with no proprietary or literature-baseline comparison, every result in this project is Qwen-internal, and the human-evaluation validation package, though fully built, was not completed in time to validate the automatic metric it was designed to check.

8.2 Hindsight and Validity

With hindsight, three process changes would likely have strengthened this project most. First, environment and infrastructure verification (GPU compute, tokenisation limits) should have been built as an automated pre-flight check run before every experiment, rather than discovered reactively after a run failed or silently misbehaved — both incidents in Chapter 5 were caught by manual inspection prompted by suspicion, not by a standing check that would catch a similar issue automatically next time. Second, even a lightweight 2–3-point hyperparameter sensitivity check, short of a full search, would have let RQ1's null result be interpreted with more confidence as a genuine ceiling effect rather than a possible artefact of under-training. Third, the human-evaluation package should have been scheduled earlier in the project timeline, budgeted as its own milestone with dedicated time set aside, rather than treated as a final step to complete once every other experiment was finished — a sequencing choice that is the direct reason it remains incomplete.

On validity specifically: internal validity is improved, but not fully resolved, by the controlled RQ1/RQ2 ablation design and the complete-container-style seen/unseen split constructed before any fine-tuning run. Construct validity is limited by the fuzzy-F1 metric's own design choices (Chapter 3) — a metric built to be lenient toward paraphrase is also, by construction, somewhat more forgiving of genuine near-misses, a trade-off made deliberately but worth naming. External validity is bounded by a single clinical corpus and a single backbone model family; nothing in this report should be read as a claim about LLM-based extraction in general.

8.3 Responsible AI, Fairness and Environmental Limits

Bias. The benchmark's sub-specialty distribution is imbalanced (Table 4, Chapter 5): general medicine accounts for 33.3% of examples, while the least-represented specialties (oncology, urology) each account for around 2%. Any model evaluated against this benchmark is therefore assessed — and, if trained on similarly distributed data, will perform — unevenly across specialties. This is compounded by the fact that the initial annotation and self-verification steps share the same underlying model family: a systematic bias in how that model perceives a given specialty's terminology risks being encoded into the gold labels themselves, rather than caught by a genuinely independent check.

Privacy. The MTS-Dialog corpus is a publicly released research dataset not drawn from identifiable real patient encounters. Any future deployment of this project's pipeline on real clinical transcripts would require dedicated de-identification, regulatory compliance, and institutional ethical approval, none of which is in scope for, or satisfied by, this academic project.

Fairness. RQ4's measured seen-versus-unseen performance gap is directly a fairness concern if any system built on this methodology were deployed operationally: patients whose conversations fall into an under-represented or entirely unseen specialty would systematically receive lower-quality automated extraction than patients in well-represented specialties, with no indication to the clinician unless the system's confidence or specialty coverage is surfaced explicitly.

Transparency. Hallucination rate is reported as a first-class metric throughout this project, not a secondary diagnostic, precisely because a syntactically well-formed structured output — particularly in the JSON and dataclass formats — can appear trustworthy while containing fabricated content. Even

the best-performing format still hallucinates 7.5–9.7% of extracted facts; this residual rate is reported explicitly rather than obscured behind an aggregate accuracy figure.

Accountability and human oversight. Given the hallucination rates and the seen/unseen generalisation gap documented in Chapter 6, this project's explicit position is that the resulting model and pipeline are appropriate only as a clinician-facing drafting aid, with mandatory human review before any extracted fact enters a patient record or informs a clinical decision. No claim is made, or should be inferred, that this system is suitable for autonomous clinical documentation or decision-making without a human clinician in the loop.

Environmental limits. All experiments were run on a single local 8 GB laptop GPU rather than a shared cluster, and every training run's duration (approximately 42 minutes for RQ1, 109 minutes for RQ2) is reported in Chapter 5 for transparency. No comparative energy-use claim is made, since exact power draw was not instrumented; this is named as a limitation rather than estimated.

8.4 AI Assistance and Student Responsibility

In the interest of the same transparency this report applies to its technical claims, the role of AI assistance in preparing this Supporting Report and the accompanying Research Paper is disclosed explicitly here, rather than left unstated.

AI assistance (Claude, accessed via Claude Code, Anthropic) was used to search, read, and cross-reference the project's own existing notebooks, scripts, saved evaluation CSVs, prediction JSON files, and prior draft reports; to organise that verified content into the chapter structure this report and the Research Paper follow; and to draft, format, and typeset both documents in Word, including the tables, in-text citations, and the two diagrams in this report (Figures 1 and 2, generated with matplotlib from the project's own recorded pipeline sequence). It was not used to run any experiment, train any model, generate any evaluation result, or invent any citation: every number in this report and the Research Paper traces to a specific artefact already produced by the student's own prior notebook and script runs (Appendix A), and every reference in the References list below is one of the ten sources the project's own literature review already cites.

This distinction matters and is stated plainly: AI assistance here organised and expressed evidence the student had already generated; it did not generate the evidence itself, and it did not exercise scientific judgement about which findings to report or how to interpret them — those interpretive choices (for example, treating RQ1's result as a genuine null finding rather than reframing it, and including Section 6.2's case study despite it showing a residual annotation error) were directed by the student. The student remains responsible for reviewing this report's content in full, verifying its claims against the underlying repository before submission, and understanding the material well enough to defend every number, methodological choice, and limitation named here under questioning in the recorded presentation.

Chapter 9 — Conclusion and Future Work

9.1 Conclusion

This report has documented, in full technical depth, a study of how an open-source large language model can be adapted for structured extraction from clinical conversation. A 798-example, human-reviewed, difficulty- and specialty-stratified benchmark was constructed from the MTS-Dialog corpus using a two-stage quality-filtering process — one strategy trialled and abandoned, one adopted — followed by full human review. Against this benchmark, four research questions were tested under a

mathematically formalised, identical scoring standard (Chapter 3). Naive LoRA fine-tuning on 250 examples produced a null result relative to prompting; embedding explicit annotation guidelines in the fine-tuning prompt produced a modest, real, positive result at negligible additional cost; JSON function-calling output decisively minimised hallucination against natural-language and Python-dataclass alternatives on every metric measured; and a measurable but partly artefactual generalisation gap was found between clinical specialties represented during fine-tuning and specialties held out entirely.

The project's single largest limitation is scope: a single clinical domain, no proprietary or literature-baseline comparison, and an unsearched hyperparameter space together mean these findings should be read as domain- and configuration-specific rather than general claims about LoRA fine-tuning, guideline-conditioning, or output-format design. Within that scope, the project delivers what it set out to: a reusable benchmark and evaluation harness, an honest empirical account — including a genuine null result reported rather than concealed — and a set of concrete, falsifiable directions for the work this project's own compute and time budget did not permit completing.

9.2 Future Work

Statistical validation: compute bootstrap confidence intervals or a paired significance test over the 89-item unseen split for the RQ1 and RQ2 comparisons, to establish whether the observed deltas exceed sampling noise.

Hyperparameter and data-scale ablation: systematically vary training-set size beyond 250 examples and epoch count, to test the leading hypothesis that RQ1's null result reflects under-training rather than a genuine ceiling effect.

Multi-domain extension: extend the benchmark-construction pipeline (Chapter 4) to customer-support and e-commerce dialogue, restoring the original plan's multi-domain scope and testing whether RQ1–RQ4's findings replicate outside the clinical domain.

Proprietary and literature baselines: run at least a partial GPT-4 zero-shot comparison on the 89-item unseen split, and either reproduce or cite comparable published numbers for InstructUIE, GoLLIE 7B, and FnCTOD-LLaMA2-13B.

Complete the human-evaluation package: finish rating the stratified 60-item sample and compute Pearson/Spearman/Kendall correlation against the automatic fuzzy-F1 metric, to validate — or correct — the metric design used throughout this project.

Fuller GoLLIE-style regularisation for RQ2: test whether class-order shuffling, class dropout, and guideline paraphrasing close the gap between this project's +2.3-point result and GoLLIE's own +13-point result, isolating how much of that gap is attributable to regularisation technique versus data and compute scale.

Mechanistic investigation of RQ3's few-shot/hallucination interaction: run controlled experiments varying output-format constraint independently of example count, to establish whether the effect observed in Section 6.6 is a general property of in-context examples or specific to less-constrained output grammars.

Two questions raised by this work remain genuinely open. First, whether RQ1's null result reflects a fixable data/compute constraint or a real ceiling effect in strongly instruction-tuned models cannot be distinguished without the data-scale ablation proposed above. Second, the precise mechanism behind RQ3's few-shot-increases-hallucination effect for less-constrained output formats was not established within this project's scope. These directions extend the evidence rather than rewrite it; the RQ1 null result, the RQ4 generalisation gap, and the RQ3 output-format effect remain the three findings most worth carrying into any next iteration of this work.

References

The following ten references form the verified corpus used throughout this report and the accompanying Research Paper. Numerical citations in the chapters above refer to this list; no additional or unverified source is introduced.

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Appendix A — Evidence Index

This index links each major claim made in this report to the primary artefact it was computed or drawn from, and states the interpretation boundary retained in the prose above.

Table 15. Primary evidence map and interpretation boundaries.

Report claim	Primary evidence artefact	Boundary retained
Benchmark composition (798, seen/unseen, tiers)	benchmark_final_split.json	Locked before any fine-tuning run; reused unchanged across RQ1/RQ2/RQ4
NLI-filtering flag rate (717/1,687)	flagged_for_review.json	Abandoned method; retained on disk as negative evidence, not deleted
Self-verification flag rate (275/1,687)	flagged_for_review_v2.json	Adopted method; every flag subsequently human-reviewed
Baseline zero-/few-shot per-field F1	evaluation_results_zeroshot.csv; evaluation_results_fewshot.csv	All 798 items; parse failures reported separately, not folded into F1
RQ1 fine-tuning comparison	finetune_comparison_summary.csv	89-item unseen split only; not the full 798
RQ2 guideline ablation	rq2_guidelines_comparison_summary.csv	Same 89-item split; guideline text is the sole independent variable
RQ3 format comparison	rq3_format_comparison_summary.csv	All 798 items, zero- and few-shot; three independent parsers
RQ4 tier / seen-unseen breakdown	RQ_Results_Summary.ipynb aggregate cells	Zero-shot condition only; manual 30-item review is a separate, smaller sample
Model-size comparison	model_size_comparison_summary.csv	Supplementary; not part of the original four research questions
Environment failures and fixes	Session verification (GPU smoke test; token length audit)	Reported as diagnosed-and-fixed, not merely noted
Human-evaluation package status	human_eval_rating_sheet.xlsx; human_eval_answer_key.csv; score_human_eval.py	Built but not completed at time of writing; disclosed as such throughout
Live demo interface and behaviour	gradio_demo.py; screenshot in Figure 9	Presentationally only; scores shown are computed by the same common_extraction.py module used for every reported result, not a separate metric

Source: Repository files named in each row; this table is itself a navigation aid, not a new source of evidence.

Appendix B — Selected Reproducibility Notes

The project repository remains the source of code, saved artefacts, and detailed logs; this appendix deliberately does not reproduce code listings or verbose per-run output. Every scoring script in the repository imports `common_extraction.py` so that JSON, natural-language, and dataclass outputs are scored by identical logic (Chapter 3); every fine-tuning run fixes random seed 42 for its training subsample; and every CSV consumed by `RQ_Results_Summary.ipynb` is read unmodified, with no value hand-edited after generation. The fixed Research Paper submitted alongside this report is not duplicated here and remains its own, separately formatted document in the required IEEE conference template.

Review points for a reader checking this report against the repository are limited to: confirm that `benchmark_final_split.json`'s stratification counts match Table 3 and Table 4 (Chapter 5); confirm that the RQ1 and RQ2 fine-tuning runs share an identical training subsample and differ only in prompt text; confirm that the human-evaluation package is described as built-but-incomplete, not completed; and confirm that this report's AI disclosure (Section 8.4) matches